Sri Lanka Institute of Information Technology



IT3091 – Machine Learning

Year 3, Semester 1 - 2026

Decision Log

EduPredict: Student Performance Prediction
Guided Data Track – Education (Student Performance)

Item	Details
Group	2026-AI-33
Campus	Malabe
Document	Decision log
Track / Dataset	Guided Data Track (code 3: Education); UCI Student Performance, Mathematics
Date	12 October 2026
Repository	https://github.com/SupunThk/EduPredict

Table P.1: Document submission metadata

Group Members

IT Number	Name	Member	Role
IT24102032	Thinila N.A.D.I.D.	Member 1	Problem Framing & Data Understanding
IT24102008	Withanan W.Y.P.	Member 2	Preprocessing & Feature Engineering
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Table P.2: Group 2026-AI-33 member details

Table of Contents

1. Introduction....3
    1.1 Purpose....3
    1.2 Status meanings....3
2. Problem Framing Decisions....4
3. Data and Preprocessing Decisions....5
4. Modelling and Evaluation Decisions....6
5. Responsible Use and Open Decisions....7

List of Tables

Table P.1: Document submission metadata....1
Table P.2: Group 2026-AI-33 member details....1
Table 1.1: Decision index....3
Table 2.1: Problem framing decisions....4
Table 3.1: Data and preprocessing decisions....5
Table 4.1: Modelling and evaluation decisions....6
Table 5.1: Responsible use and open decisions....7

1. Introduction

1.1 Purpose

This log records the current project decisions and their evidence as of 12 October 2026. It preserves the original D01 to D14 identifiers, consolidates the duplicate dataset-scope entry into D15, and records the updated zero-grade decision in D16. Completed experiments settle modelling choices; operational availability and one source-code ambiguity remain open.

1.2 Status meanings

Status meanings: Decided records an adopted project choice; Completed records a finished evaluation; Open requires further evidence. Entries describe project decisions rather than approval by a school or all team members.

ID	Topic	Status
D01	Primary task	Decided
D02	Secondary lens	Decided
D03	Prediction unit	Decided
D04	Stakeholders	Decided
D05	Use of prior grades	Decided
D06	Missing values	Decided
D07	Encoding	Decided
D08	Scaling	Decided
D09	Benchmarks	Decided
D10	Candidate models	Decided
D11	Metrics	Decided
D12	Selection	Decided
D13	Threshold	Decided
D14	Flag terminology	Decided
D15	Subject scope	Decided
D16	Zero final grades	Decided
D17	Data split	Decided
D18	Cross validation	Decided
D19	Absence transformation	Decided
D20	Reusable pipelines	Decided
D21	Final test evaluation	Completed
D22	Support and responsible use	Decided
D23	Dictionary discrepancy	Open
D24	Prediction timing and adoption	Open

Table 1.1: Decision index

2. Problem Framing Decisions

ID	Decision	Rationale	Evidence
D01	**Primary task. Decided.** Regression rather than a standalone classification task.	Estimate continuous G3 while retaining grade-error information. G3 is integer-valued on a 0–20 scale.	Business Problem; Notebook 03
D02	**Secondary lens. Decided.** Derive the risk flag from the regressor rather than train a separate classifier.	Use the same predicted grade for staff review and classification metrics.	Notebook 04 sections 2 and 4
D03	**Prediction unit. Decided.** Individual student record rather than class or school.	Each Mathematics CSV row represents one student observation.	data/student-mat.csv
D04	**Stakeholders. Decided.** Teachers and academic support staff are the primary users; students are affected stakeholders.	Staff retain responsibility for support decisions. The two-school sample does not validate use at every institution.	Business Problem; Notebook 04 recommendations
D15	**Subject scope. Decided.** Use Mathematics only, rather than Portuguese or combined subjects.	Keep one subject and one student-record population. The source file is student-mat.csv with 395 records.	data/student-mat.csv

Table 2.1: Problem framing decisions

3. Data and Preprocessing Decisions

ID	Decision	Rationale	Evidence
D05	**Use of prior grades. Decided.** Compare two scenarios: main excludes G1/G2; later-stage comparison includes both.	G1/G2 can be used only once available. Correlation with G3 alone does not establish leakage.	src/ preprocessing.py; five model notebooks
D06	**Missing values. Decided.** No imputation or row removal for missingness in the current data.	The Mathematics file has no missing cells. Missing or unexpected binary values are rejected; future-data handling still requires design.	395-row CSV audit; src/preprocessing.py
D07	**Encoding. Decided.** Map 13 binary features, one-hot encode four nominal features and pass ten ordered codes as numeric inputs.	Model notebooks drop the first one-hot level and ignore unknown nominal categories. Treating ordered codes numerically assumes useful spacing; this is a modelling approximation.	src/ preprocessing.py; model notebooks
D08	**Scaling. Decided.** Standardize numerical and ordered inputs for Linear Regression and Ridge; use no scaling for tree models.	Fit scaling within each training fold. Binary maps and nominal encodings are kept separate.	Five model notebooks; src/preprocessing.py
D16	**Zero final grades. Decided.** Retain all 38 recorded zeros, replacing the earlier positive-grade-only scope.	Zeros are not proven missing values or errors. Earlier 357-student results are not directly comparable; their exported files were removed. Scope changed 11 October 2026.	docs/notes/ zero_grade_scope_ change.md; archived experiment records
D17	**Data split. Decided.** Use a common 316/79 split with test_size=0.2 and random_state=42.	Keep exact row IDs consistent across models and scenarios; no student belongs to both sets.	src/model_utils.py split; selected metadata
D19	**Absence transformation. Decided.** Search original versus log1p absences, rather than choose from full-data skew alone.	The selected main forest uses original absences; the later-stage forest uses log absences. No post-test transformation change is made.	Both selected model metadata files
D20	**Reusable pipelines. Decided.** Share preprocessing in src/preprocessing.py and save complete fitted pipelines.	Keep preprocessing and estimator together; verify data/code/model hashes, versions and row IDs before comparison and evaluation.	src/ preprocessing.py; models/*.joblib; metadata

Table 3.1: Data and preprocessing decisions

4. Modelling and Evaluation Decisions

ID	Decision	Rationale	Evidence
D09	**Benchmarks. Decided.** Use training-derived mean and median constants, plus a G2-unchanged later-stage test reference.	Mean and median test a model against simple predictions; G2 checks whether later-stage complexity adds value. Linear Regression remains a candidate.	Model notebooks; Notebook 04
D10	**Candidate models. Decided.** Compare Linear Regression, Ridge, Decision Tree, Random Forest and Gradient Boosting.	Give each model its own initial configuration, justified search, diagnostics and two feature scenarios.	notebooks/03a through 03e
D11	**Metrics. Decided.** Report MAE, RMSE and R^2; also accuracy, precision, recall, F1, specificity and confusion counts for flags.	Grade error and grade-10 classification answer different questions. Undefined precision is missing. Report fold means/std separately from pooled rates.	src/model_utils.py; Notebook 04
D12	**Selection. Decided.** Select the lowest mean training-CV MAE separately for each scenario; Random Forest wins both.	Supporting metrics explain tradeoffs without changing the original selection rule retrospectively. Training-CV MAE: 2.964 without prior grades, 0.938 with prior grades.	Notebook 03; selected model metadata
D13	**Threshold. Decided.** Use continuous predicted G3 < 10; observed G3 < 10 defines the positive class.	Keep the project threshold fixed. No threshold search is performed on the held-out set.	Notebook 04; src/model_utils.py
D14	**Flag terminology. Decided.** Use potentially at risk and not flagged.	A prediction is uncertain; absence of a flag does not mean a student needs no help.	Business Problem; Notebook 04
D18	**Cross validation. Decided.** Use five shuffled training folds with random_state=42 and MAE-refitted grid searches.	Fit preprocessing within folds. Selected-setting scores reuse search folds and may be optimistic; they are not nested-CV estimates.	Model notebooks; training CV tables in the model notebooks
D21	**Final test evaluation. Completed.** Evaluate the two frozen selected forests on the same 79 test rows without refitting or retuning.	Main MAE 3.095, RMSE 3.871, R^2 0.269; later-stage MAE 1.170, RMSE 1.975, R^2 0.810. Test results are now observed and cannot guide new tuning.	Notebook 04 setup and final test tables

Table 4.1: Modelling and evaluation decisions

5. Responsible Use and Open Decisions

ID	Decision	Rationale	Evidence
D22	**Support and responsible use. Decided.** Use staff review and supportive recommendations; do not automate consequential student decisions.	Main flags miss 17 of 27 at-risk test students. Later-stage perfect observed recall does not guarantee future recall. Feature effects are not causal intervention evidence.	Notebook 04; final recommendations report
D23	**Dictionary discrepancy. Open.** Keep failures codes as observed 0,1,2,3; do not assert a confirmed top-coded bucket meaning.	data/student.txt describes failures inconsistently with the CSV. Confirm the original coding if it affects interpretation; do not silently recode. Distinguish documented and observed ranges.	data/student.txt; Mathematics CSV audit
D24	**Prediction timing and adoption. Open.** Leave operational timing, input availability and school acceptance criteria unresolved.	Verify the collection window for every input, especially absences; validate on a new cohort and review privacy and subgroup performance before a pilot.	Notebook 04 limitations; Business Problem

Table 5.1: Responsible use and open decisions